

Reactive path planning for autonomous sailboat

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Abstract—Among Autonomous Surface Vehicles (ASV), sailboat robots could be an efficient solution for long term missions and semi-persistent presence in the oceans since they rely on renewable energy like solar and wind energies. However, the steering of such vehicles presents an inherent difficulty since the thrust force heavily rely on the wind (speed and direction) and sail angle as well as on the underlying complex dynamic properties of the ship, mainly due to the aero and hydrodynamic forces acting on the sails and on the hull. In this paper, a new reactive path planning method is proposed, which takes into account the so called no-go zones and turns them into virtual obstacles. A potential field algorithm is then applied to drive the sailboat towards the goal while avoiding obstacles even when the goal is located directly downwind or upwind. Simulations, using a boat specific dynamic simulator, show that the proposed method can successfully drive an autonomous boat to a set of predefined way-points under time varying wind conditions. This method is computationally efficient and is suitable for on-board real-time implementation.

I. INTRODUCTION

During the recent years, a growing activity concerning autonomous surface vehicles have been noted. Applications for such vehicles include port protection, mines countermeasure, reconnaissance and surveillance mission [1]–[3]. One of the probably most important reason that can explain this growing activity is the need for better environmental monitoring. Meteorological and ecological studies in particular need better characterization of the global Earth's processes and consequently requires a broad spectrum of observational tools to understand the complex interactions between the oceans and the atmosphere.

Ocean sensing have been typically done with satellites, airplanes, buoys, research vessels or ship of opportunity. However, satellites and airplanes are limited by cloud cover, temporal/geographical coverage, as well as spatial resolution. Manned research vessels are expensive to deploy and usually suffer from a weak of availability. Spatial sampling is impossible with moored buoys and they cannot be self-deployed to specific areas of interest. Over the last decades, the growing needs of the oceanographic community for in-situ measurements and complementary observing systems have stimulated the robotic community of academic researchers as well as private companies. Thus, considerable progress have been observed in the development and use of autonomous marine platforms such as drifters, gliders, autonomous underwater vehicles (AUV), and autonomous surface vehicles (ASV).

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Each of these platforms provides various capabilities for payload, endurance, range, communication, mobility, and autonomy. In the recent years, numerous ASV have been developed for bathymetry data recording in shallow water [4], monitoring of various marine environmental data either alone [5]–[7] or as part of a sensor network [8]–[12]. All of these aforementioned vehicles are conventional electric powered systems and suffer from a lack of autonomy that restrains their usage to short term missions. Sailboat robots could be an efficient solution for long term missions and semi-persistent presence in the oceans since they rely on renewable energy like solar and wind energies [13], [14]. However, steering such vehicles presents an inherent difficulty since the thrust force heavily rely on the wind (speed and direction) and sail angle as well as on the underlying complex dynamic properties of the ship, mainly due to the aero and hydrodynamic forces acting on the boat.

The paper is organized as follows. First, a dynamic model of our sailboat prototype will be presented and used to derive a numerical simulator that will be validated using experimental data observed from a towing tank. Second, a routing strategy for the reactive navigation of autonomous sailboats is proposed. Since accurate weather forecasts are not available for short distances and periods, our algorithm takes into account only local and instantaneous wind conditions to compute an optimum heading for the boat. The path planning algorithm will react to changes of the wind conditions as well as detected obstacles in real-time by recalculating the heading periodically, similarly to what a human sailor does on a short regatta. This method have been successfully tested in under various wind conditions and have been validated in simulation using the aforementioned numerical simulator.

II. SAILBOAT MODELING

In this section the dynamic model of our prototype is presented. Both aero and hydrodynamic forces acting on the sailboat are taken into account to derive a realistic and computationally efficient numerical simulator.

A. Aero and Hydrodynamic Forces

The energy of the wind is transmitted to the system through the sails. The force acting on the sail can be expressed in terms of lift and drag components, both applied in a center of effort. The magnitudes of these forces are related to the lift coefficient C_l and the drag coefficient C_d and can be calculated, in a sail reference frame, by the formulas of fluid mechanics (1).

$$\begin{cases} F_D &= \frac{1}{2}\rho SV^2 C_d \\ F_L &= \frac{1}{2}\rho SV^2 C_l \end{cases} \quad (1)$$

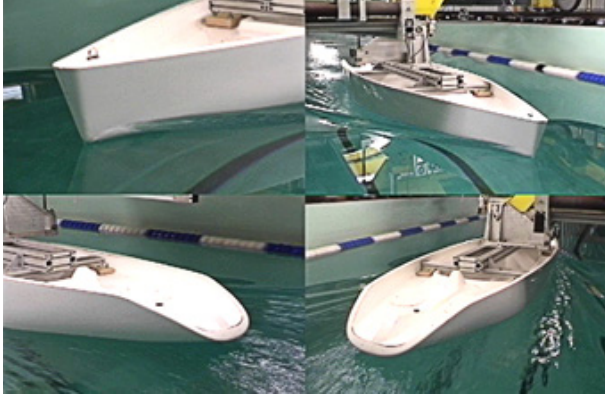


Fig. 1. Hull attached to a 6 components force-torque sensor in the towing tank of Ecole Centrale de Nantes

where ρ is the air density, S is the sail area and V is the apparent wind speed relative to the sail.

Since the rudder and the keel are airfoil-shaped bodies moving through a fluid, the hydrodynamic forces acting on these elements can also be computed using an expression similar to (1) and replacing ρ by water density, V by the relative water velocity and by using the specific lift and drag coefficients for the rudder and the keel.

The total hydrodynamic resistance R_{Hydro} acting on the hull can be written as (2):

$$R_{Hydro} = R_{Residuary} + R_{Friction} + R_{Leeway} + R_{Heel} \quad (2)$$

where $R_{Residuary}$ is the residuary resistance (mainly due to the waves created by the shape of the hull), $R_{Friction}$ is the resistance induced by the fluid on the hull, R_{Leeway} is the resistance induced by the leeway (angle between the boat main axis and the velocity vector) and R_{Heel} is the heel (roll angle) induced resistance [15].

Since the geometric parameters of the hull are compatibles with the valid interval of the Gerritsma's model [16], [17], this model is used to compute $R_{Residuary}$.

The other components $R_{Friction}$, R_{Leeway} and R_{Heel} of the hydrodynamic resistance of the hull are then experimentally identified using the towing tank (150m $\times$ 5m $\times$ 3m) of *Ecole Centrale de Nantes*. The hull, without keel and rudder, was mounted on a six DoF force-torque sensor with non restricted heave and sway (Fig. 1) and the resulting forces and torques have been recorded under various traction conditions, with and without heel and leeway for a range of ship velocities compatible with a Froude number $F_N \leq 0.45$.

Based on the data recorded during the tests, a set of interpolated functions has been derived to model the total hydrodynamic resistance.

B. Motion Equations and Simulator

The sailboat motion equations can be written using the Newton-Euler's law in the reference frame R_B , as (3):

$$\begin{cases} m \left(\frac{d\vec{V}_{B/R_0}}{dt} \right)_{/R_B} = \vec{F}_{/R_B} - m(\vec{\Omega}_{R_B/R_0} \wedge \vec{V}_{B/R_0}) \\ \Pi_{B/G} \cdot \vec{\Omega}_{R_B/R_0} = \vec{M}_G - \vec{\Omega}_{R_B/R_0} \wedge \Pi_{B/G} \cdot \vec{\Omega}_{R_B/R_0} \end{cases} \quad (3)$$

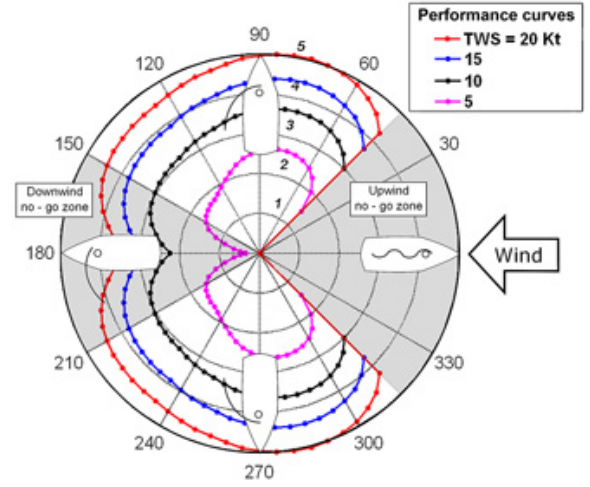


Fig. 2. Speed polar diagram of our sailboat

$\vec{F}_{/R_B}$:	total force acting on the sailboat.
$\vec{M}_G$:	total torque acting on the sailboat
m :	mass of the sailboat
$\Pi_{B/G}$:	inertial matrix of the sailboat
with: $\vec{V}_{B/R_0}$:	velocity vector of CoG in R_0
$\vec{\Omega}_{R_B/R_0}$:	angular velocity vector in R_0
R_0 :	Earth-fixed inertial frame
R_B :	body-fixed reference frame.

From this analytical description of the dynamics of our sailboat, a numerical simulator has been built, which integrates the differential equations of motion using a classical Runge-Kunta algorithm. This numerical simulator can then be used to simulate the behaviour of our prototype under different environmental conditions (see section IV).

The numerical model can also be used to compute the speed polar diagram of the sailboat (see Fig. 2). These performance curves correspond to the steady-state maximum velocity of the vehicle for a given wind speed and a given wind direction and for headings φ relative to the wind direction varying from 45° to 315° . To keep a constant heading with respect to the wind until the simulation reaches the steady-state for the vehicle velocity, a PID controller on the rudder and on the boom has been used.

For wind speeds varying from 5 to 20 knots, speed profiles look similar. A so-called *upwind no-go zone* is discarded for $-45^\circ \leq \varphi \leq 45^\circ$. Another "touchy" sailing zone, called *downwind no-go zone*, is exhibited for $150^\circ \leq \varphi \leq 210^\circ$. In fact the boat is able to navigate in this region, but for safety reasons (instability) these headings will also be avoided in practice. The performance curves will also be used to trim the sail angle in real-time to reach a given reference in terms of vehicle velocity and heading for given wind speed and direction. The description of this sail trimming algorithm is out of the scope of this article.

A high potential will be affected to these upwind and downwind sailing regions in the path planner presented in the next section.

III. PATH PLANNING

In the recent years, numerous methods have been proposed for the path planning of autonomous sailboat robots : based on logic rules [18], [19], state machine [20]–[22], neural control [23] or ray-tracing techniques [24].

Potential fields methods, due to Khatib [25] and Krogh [26], have been successfully employed for robot control and motion planning of mobile robots (see for example [27]–[30] among many others).

In artificial potential fields methods, movements of the robot (represented as a particle) are governed by a field which is usually composed of two components, an attractive potential drawing the robot towards the goal and a repulsive potential pushing the robot away from obstacles. The originality of our method is to build a so-called local potential around the boat location to take upwind and downwind sailing constraints into account. Moreover, an hysteresis potential relative to the cost of tacking and gybing is also proposed, which makes our method easy to tune according to the vehicle specifications.

The main drawback with potential field techniques is their susceptibility to local minima. However, typical marine environments are sparse and local minima in this case are not an issue. Besides, the reliability of our path planning method will benefit from coupling the high level potential based path planner with a low level realistic simulator derived from the sailboat model identified in section II.

A. Potential field approach

Our potential field approach is decomposed in two parts, a global potential field and a local potential field. The global potential field is relative to the goal point and to the obstacles. The local potential field is relative to the wind direction.

1) *Global potential*: The global potential is built on the all map once for all at the beginning of the planning process. It aims at attracting the vehicle towards the goal point and repulsing the vehicle outwards the obstacles.

In order to attract the vehicle towards the goal point P_{goal} a linear potential P_g is built for every point P of the map as follows:

$$P_g = G_g \cdot \text{dist}(P, P_{goal})$$

where G_g is the desired attractive gradient and $\text{dist}(P, P_{goal})$ is the Euclidean distance between P and P_{goal} . Since G_g is a constant, the potential P_g looks like a cone centered on P_{goal} , see Fig. 3.

A repulsive potential P_o relative to the obstacles located in P_{obst} is built for every point P of the map as follows:

$$P_o = \frac{k}{\text{dist}(P, P_{obst})}$$

where k is a tunable scalar. This potential tends to infinite as we come close to the obstacles, which prevents the vehicle to go through the obstacles. The position of each obstacle can be known in advance or can be updated in real time by on board navigation sensors.

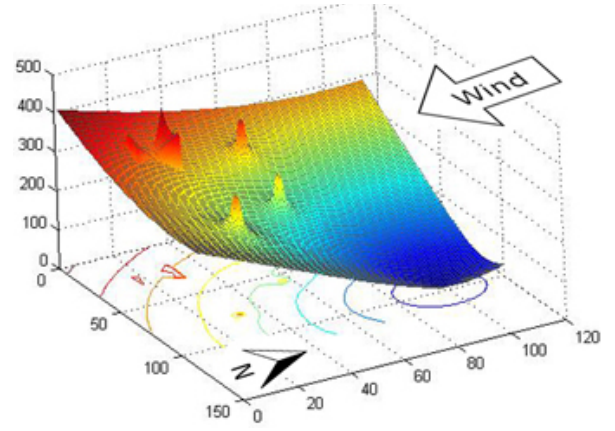


Fig. 3. Illustration of the overall potential relative to three obstacles and to a goal point located at the North-East corner of the map

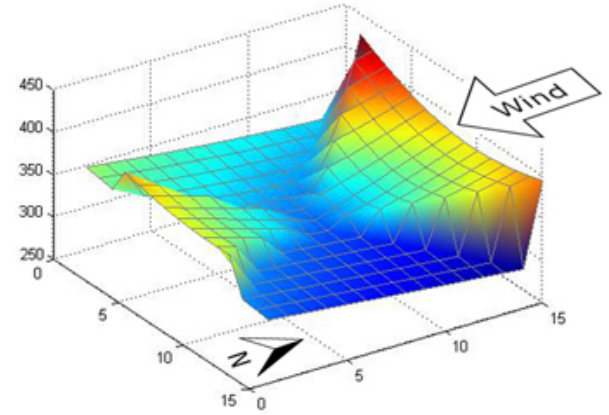


Fig. 4. Close-up on the local potential around the boat. In this case, the upwind no-go zone is pointing up because the wind is coming from North

2) *Local potential*: The local potential is built in the neighborhood of the boat location. It is rebuilt after each displacement of the boat according to the wind direction and the latest tack, see Fig. 4.

In order to take the upwind no-go zone into account a so-called upwind potential P_{up} is built dynamically for each point P_w inside a window (size W_{size}) centered on the boat location P . The magnitude of P_{up} is a function of the angle ϕ between the direction of $\vec{PP_w}$ and the wind direction TWA . It is linear inside the upwind no-go zone sector defined by the angle ϕ_{up} (see Fig. 5) and null outside:

$$\begin{cases} P_{up} = G_{up} \cdot \text{dist}(P_w, P) & \text{if } 0 < |\phi| < \phi_{up} \\ P_{up} = 0 & \text{elsewhere} \end{cases}$$

where G_{up} is the desired gradient inside the upwind no-go zone sector.

In order to take the downwind no-go zone into account a so-called downwind potential P_{down} is built dynamically for each point P_w inside the same window centered on the boat location P . The magnitude of P_{down} is a function of the angle between the boat direction ϕ and the wind direction.

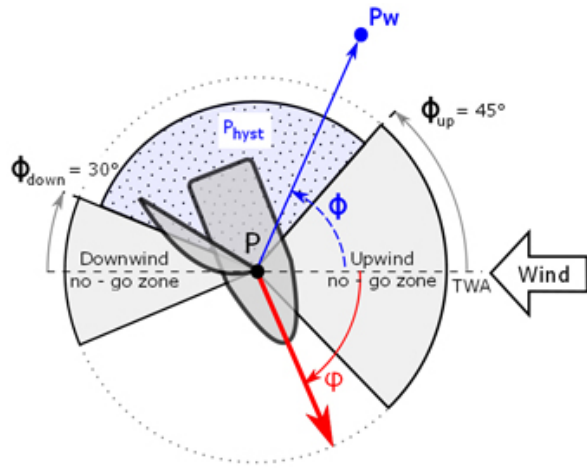


Fig. 5. Scheme of the upwind and downwind no-go zones. The dotted region corresponds to the region where $P_h \neq 0$

It is linear inside the downwind no-go zone sector defined by the angle ϕ_{down} (see Fig. 5) and null outside:

$$\begin{cases} P_{down} = G_{down}.dist(P_w, P) & \text{if } 0 < |\phi - \pi| < \phi_{down} \\ P_{down} = 0 & \text{elsewhere} \end{cases}$$

where G_{down} is the desired gradient inside the downwind no-go zone sector.

In order to take the cost of gybing and tacking into account a so-called hysteresis potential P_h is built dynamically for each point P_w inside the window centered on the boat location P . The magnitude of P_h is a function of the angle ϕ and the latest tack. For instance, if the boat navigates with the wind coming from its left side, P_h will be defined at the right side of the wind direction to prevent the boat to tack or gybe too frequently (see Fig. 5). In this example, P_h is defined as follows:

$$\begin{cases} P_h = G_h.dist(P_w, P) & \text{if } \phi_{up} < \phi < \pi - \phi_{down} \\ P_h = 0 & \text{elsewhere} \end{cases}$$

where G_h is the desired gradient of P_h . This way, only one parameter has to be set up to fit the cost of gybing and tacking.

3) *Gradient descent*: Once the local potential has been built and added to the global potential, a gradient descent is carried out on the overall potential $P_t = P_g + P_o + P_{up} + P_{down} + P_h$ to find the new heading for the boat. The gradient descent algorithm is stopped when the boat arrives in the vicinity of the goal point. Precisely, the stop criteria is defined as follows: $dist(P, P_{goal}) < \epsilon$, with $\epsilon = W_{size}$.

B. Simulation results

In this section, the influence of wind configurations and sailboat capabilities on trajectories will be analyzed in simulation. Realism of resulting trajectories from our reactive path planning approach will eventually be discussed.

1) *Simulated environment*: For the sake of clarity, the simulated environment is assumed to be static (obstacles are fixed) and completely known (obstacles are mapped without

uncertainty). Besides, the wind is assumed to be constant in direction as well as in velocity. However, since our local potential is incrementally built as a function of the visible wind and obstacles around the boat location, this reactive path planning approach holds for dynamic environments and can be implemented without any change on real embedded systems.

2) *Simulated trajectories*: In the following simulations, default parameters for the potential have been set up as follows: $G_g = 3$, $k = 100$, $G_{up} = 10$, $G_{down} = 5$, $G_h = 2$, $\phi_{up} = 45^\circ$, $\phi_{down} = 30^\circ$.

a) *Downwind navigation*: Two trajectories exhibited by the potential field based path planner for winds coming from South-West and from West are depicted in Fig. 6.

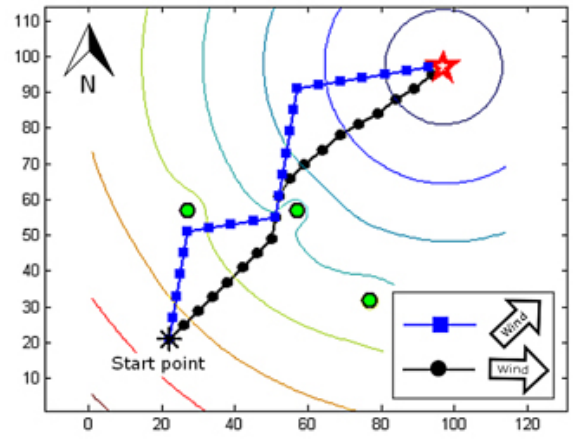


Fig. 6. Trajectories computed with winds coming from South-West and West

In case of wind coming from West, the sailboat navigates directly towards the goal point while avoiding obstacles without any difficulty. In case of wind coming from South-West, the optimal trajectory in respect with the potentials built by our planner is more complex. The boat avoids two obstacles and executes three gybes on its way to the goal point.

b) *Upwind navigation*: A trajectory solution exhibited by the potential field based path planner for a wind coming from North-East is depicted in Fig. 7. Several hysteresis potentials have been tested. The red trajectory has been computed with $G_h = 3$, the blue trajectory has been computed with $G_h = 2$ and the black trajectory has been computed with $G_h = 1$. Two, three and seven tacks have been respectively executed to reach the goal point.

c) *Conclusion*: The presented path planner provides coherent trajectories in terms of heading constraints for a sailboat. Trajectories follow admissible headings regarding the upwind and downwind no-go zones. The number of gybes and tacks may be tuned by the hysteresis potential to suit the real cost of these manoeuvres. Besides, attractiveness of the goal point and repulsiveness of obstacles are easily tunable in this potential field framework.

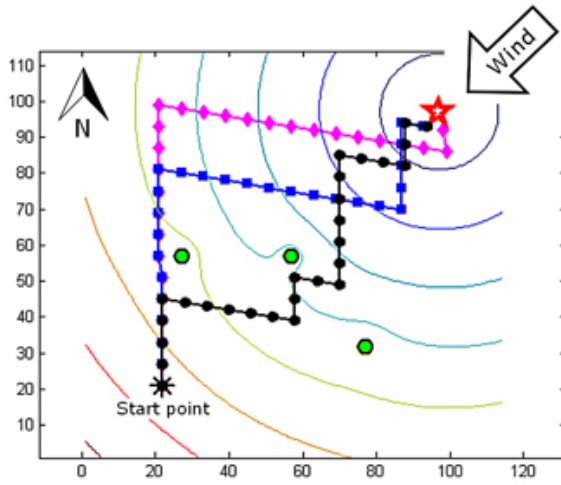


Fig. 7. Trajectories computed with the wind coming from North-East using different hysteresis potentials. In red $G_h = 3$, in blue $G_h = 2$, in black $G_h = 1$

IV. NUMERICAL PATH PLANNING VALIDATION

In this section, realism of the simulations will be improved both in terms of external environment and in terms of internal model of the boat. For this purpose, a dynamic wind field will be introduced and the dynamic model of the sailboat identified in section II will be used.

A. Realistic environment

In this section the path planner is tested using a more realistic model of a marine wind field. As depicted in Fig. 8, the wind in this path planning simulation varies from West to East along the horizontal axis. The trajectory exhibited by the potential field based path planning algorithm bends to the East as the wind turns from West to North up to a point where a tackle becomes necessary to reach the goal point.

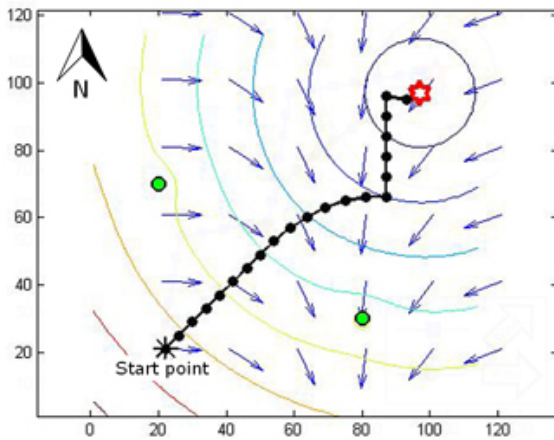


Fig. 8. An example of trajectory computed with a dynamic wind field

B. Realistic boat model

In the simulations so far the vehicle dynamics has been neglected. Resulting paths extracted from the start to the goal point may not be physically admissible for a given sailboat. Even if these trajectories can be considered from a high level point of view, introducing the sailboat dynamics would greatly improve the reliability of the overall path planning method. For this reason a numerical simulator will be coupled with our path planner to give us more realistic simulations.

1) *Simulator description:* The numerical simulator developed in section II is called as a DLL library. Inputs at time t are the attitude (roll, pitch and yaw), position (x, y, z), linear and angular speeds of the boat, the boom angle, the rudder angle, true wind angle and true wind speed. Outputs are the same variables at step $t + \delta t$, where $\delta t = 50ms$. In the following simulations the state of the boat is initialized at zero for linear and angular speeds as well as for the attitude, and the boat is initially pointing towards the goal point. The low level control loop for the boom angle and for the rudder angle is a simple PID controller that makes the heading converging to the heading reference given by the high level path planner.

2) *Results:* A comparison between a trajectory computed using only the high level path planner and a trajectory resulting from the coupling of the path planner with the low level simulator is given in Fig. 9. In this framework, where the wind comes from East, the boat begins to accelerate in the direction given by the path planner. At this stage of the simulation both trajectories are similar in terms of heading. To reach the target a jibe is theoretically necessary at some point of the trajectory, as depicted by the black curve. While easily done when neglecting the vehicle dynamics, this manoeuvre is far from immediate in practice. The inertia of the boat may impact such a turn in the trajectory, as depicted by the blue curve.

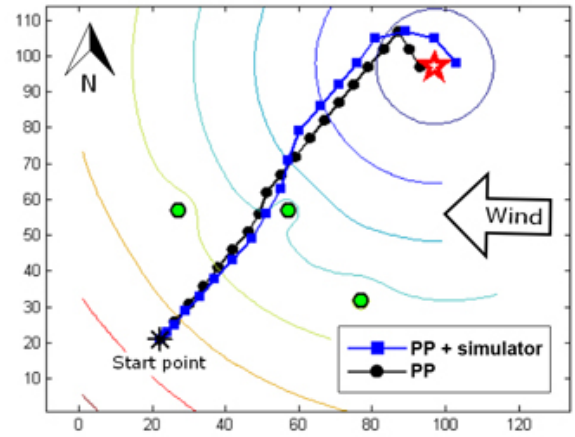


Fig. 9. Trajectories computed with the high level path planner only (PP) and by coupling the high level path planner and the low level simulator (PP + simulator)

V. CONCLUSION AND FUTURE WORKS

In this work, a reactive path planning method for the navigation of an autonomous sailboat has been presented. This path planning algorithm, based on the potential field approach, reacts to changing wind conditions as well as detected obstacles in real-time by recalculating the heading periodically. This method have been successfully tested and validated in simulation under various wind conditions, using a computational efficient numerical simulator specifically tuned for the boat that will be used in the next future.

This 3.6 m long sailboat based on a mini-J hull (a reduced scale of a 12M JI class) is currently under development, see Fig. 10. The electrical power is provided by a battery pack, charged by a wind turbine and a 0.5 m² solar panel. She will use on-board navigation sensors (IMU, GPS, anemometer and wind vane) as well as perception sensors (sonar, hydrophones and panoramic camera) for obstacle detection.



Fig. 10. Illustration of our prototype during the first trial at Nantes in France

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